9.4 Systemic Mycoses

Systemic mycoses are caused by dimorphic fungi; these infections are acquired by inhalation of spores. These primarily involve the respiratory system and are self-limiting and asymptomatic. If symptomatic, it spreads to other parts of body through circulation. These infections are caused by **true fungal pathogens**. Systemic and opportunistic infections together caused **Deep mycoses**.

The organisms have a mycelial form when grown on fungal culture and have yeast form in the tissue. The examples of systemic mycoses are Histoplasmosis, Blastomycosis.

9.4.1 Histoplasmosis

Histoplasmosis is caused by dimorphic fungus ***Histoplasma capsulatum***. The fungi live inside the cells of the reticuloendothelial system, where they grow within macrophages and giant cells. This infection is also known as **Darling's disease**.

9.4.2 Pathogenesis and Pathology

The infection with H. *capsulatum* develops when conidia or mycelial fragments are in haled and converted into yeasts in alveolar macrophages in the lungs. The oval yeast cells parasitize macrophages, which are activated by T lymphocytes resulting in localized granulomatous inflammation.

HOTS

H. *capsulatum* is dimorphic fungi - justify

9.4.3 Clinical Features

The disease is mostly asymptomatic. The development of symptom or symptomatic disease appears to depend on the intensity of exposure to conidia and cellular immune response of the host. The disease may be classified as follows.

1. Acute pulmonary Histoplasmosis – Fever, headache, chills, sweating, chest pain, cough and dyspnoea
2. Chronic pulmonary Histoplasmosis – Ulcerative lesions of the lips, mouth, nose, larynx and intestines
3. Cutaneous, mucocutaneous Histoplasmosis – Mucous lesions on skin, abdomen wall and thorax.
4. Disseminated Histoplasmosis – Fever, anoxia, anemia, leucopenia constant hepatosplenomegaly and multiple lymphadenopathies.

9.4.4 Laboratory Diagnosis

i. Samples

Specimens collected are sputum, bone marrow and lymph nodes, cutaneous and mucosal lesions and peripheral blood film.

a. Direct Examination

Thick and thin smears should be prepared from peripheral blood, bone marrow and stained with Calcofluor white, Giemsa or Wright stains.

The fungus is small, oval yeast like cells, 2–4 µm in diameter, within the mononuclear or polymorpho nuclear cells and occasionally in giant cells.

b. Fungal culture

The clinical samples is inoculated on Sabouraud dextrose agar (SDA) and

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